

SECTION 11.1

Structures and Processes of the Nervous System

Section Outcomes

In this section, you will

- **describe** the role of the nervous system in homeostasis
- **describe** and **observe** the organization of neurons into nerves
- **describe** the structure and function of a reflex arc, and **design** and **perform** an investigation to study reflex arcs
- **explain** the transmission of an action potential along a neuron and from one neuron to another, and **perform** investigations that demonstrate the structure and function of neurons
- **analyze** and **explain** the effects of various drugs on the nervous system

Key Terms

homeostasis
nervous system
central nervous system
peripheral nervous system
neurons
glial cells
nerves
reflex arcs
dendrites
cell body
axon
myelin sheath
Schwann cells
membrane potential
resting membrane potential
polarization
sodium-potassium exchange pump
depolarization
nodes of Ranvier
action potential
threshold potential
repolarization
refractory period
synapse
neuromuscular junction
neurotransmitters
acetylcholine
cholinesterase



Figure 11.1 This child is wearing the same kind of clothing that her people, the Inuit, have traditionally made and worn to protect them against the extremes of their Arctic environment. How is her nervous system adapted to enable her body to maintain homeostasis while living in a harsh climate?

The human nervous system is equipped to sense and respond to continuous change within the body and in its external environment. As you may recall from your previous studies of human systems, **homeostasis** is a state of relative stability within the body. Homeostasis is critical for survival, because the body can only survive within a narrow range of conditions. The nervous system regulates body structures and processes to maintain homeostasis despite fluctuations in the internal and external environment.

For example, the child in Figure 11.1 lives in what many people consider to be an inhospitable environment, with winter temperatures often falling to -50°C and lower. For the Arctic Inuit, maintaining a constant internal temperature, while keeping blood and heat flowing to the extremities, is crucial. Researchers have discovered that the nervous systems of people living in warmer climates act to constrict blood flow to an extremity (and thus conserve body heat) when the extremity is cooled. In Inuit people who have lived for generations in the far North, however, the nervous system fluctuates the

constriction and dilation of blood vessels to cooled extremities. This has the effect of conserving body heat, but it also allows for continued blood flow to prevent frostbite. In this way, homeostasis is maintained.



Figure 11.2 Dissections allow researchers to learn about the human body. This dissection of the human nervous system was donated to the Smithsonian Institution in 1935 from the Kirksville College of Osteopathic Medicine.

To maintain homeostasis, the human body must react to differences in temperature as well as respond to various internal and external stimuli, and it must regulate these responses. The human nervous system can regulate tens of thousands of activities simultaneously. In ways that scientists do not fully understand, it is intimately connected with human consciousness, intelligence, and creativity. The nervous system monitors and controls most body processes, from automatic functions (such as breathing) to activities that involve fine motor coordination, learning, and thought (such as playing a musical instrument). The brain and spinal cord, and the nerves that emerge from them and connect them to the rest of the body, make up the human **nervous system** (Figure 11.2).

- 1 Define homeostasis.
- 2 Explain why the nervous system is critical for maintaining homeostasis.

Organization of the Nervous System

The human nervous system is perhaps the most complex system of any organism. The human brain alone contains over 100 billion nerve cells, and each nerve cell can have up to 10 000 connections to other nerve cells. This means that a nerve impulse—an electrochemical signal—to or from the brain could travel along 10^{15} possible routes.

The nervous system has two major divisions: the central nervous system (CNS) and the peripheral nervous system (PNS). Early researchers made this distinction based on where nervous tissue was located in the body—centrally or away from the centre (peripherally). Together, the central nervous system and the peripheral nervous system control sensory input, integration, and motor output.

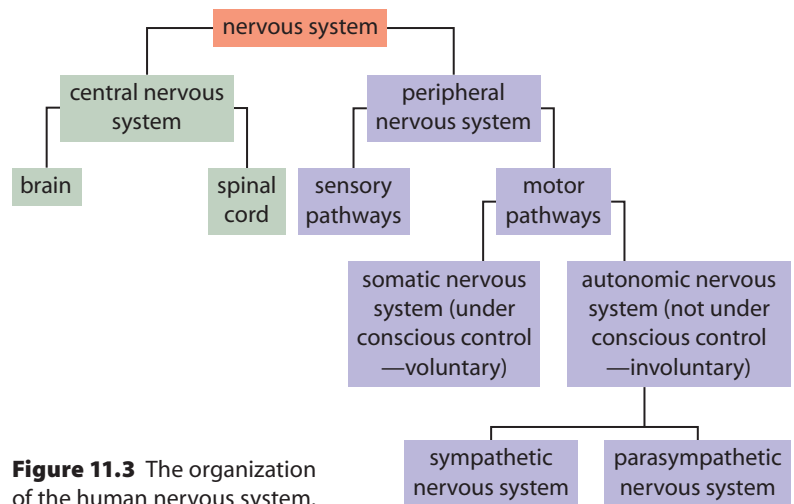


Figure 11.3 The organization of the human nervous system.

The **central nervous system**, which consists of the brain and spinal cord, integrates and processes information sent by nerves.

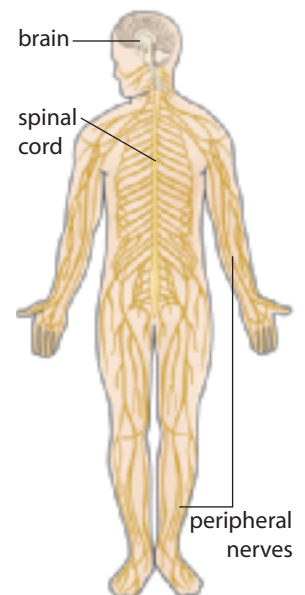
The **peripheral nervous system** includes nerves that carry sensory messages to the central nervous system and nerves that send information from the CNS to the muscles and glands.

The peripheral nervous system is further divided into the somatic system and the autonomic system.

The *somatic system* consists of sensory receptors in the head and extremities, nerves that carry sensory information to the central nervous system, and nerves that carry instructions from the central nervous system to the skeletal muscles.

The *autonomic system* controls glandular secretions and the functioning of the smooth and cardiac muscles.

The *sympathetic* and *parasympathetic* divisions of the autonomic system often work in opposition to each other to regulate the involuntary processes of the body. Involuntary processes, such as heartbeat and peristalsis, are those that do not require or involve conscious control.



- 3 Create a table to identify the different systems in the nervous system and explain the structure and function of each.

BiologyFile

FYI

Most nerves are buried deep in the body to protect them from injury, with only small extensions that reach the outer skin. One exception is the ulnar nerve at the elbow, which runs close to the surface of the skin. This is why a slight knock to your elbow sends an unpleasant tingling sensation throughout your arm!

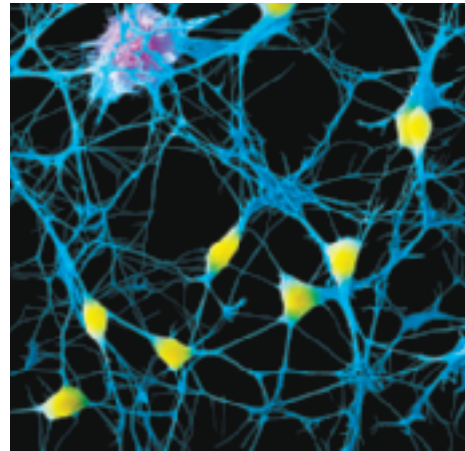
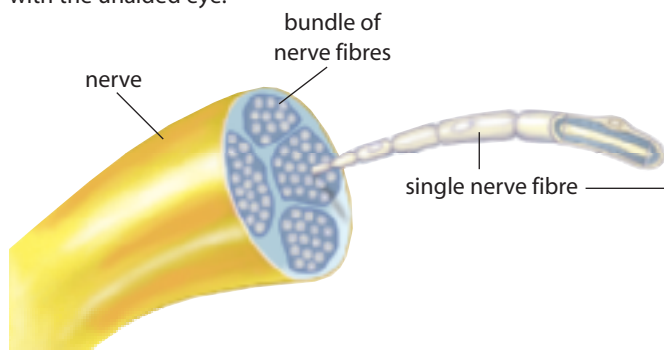
Cells of the Nervous System

The nervous system is composed of only two main types of cells: neurons and cells that support the neurons, which are called glial cells. **Neurons** are the basic structural and functional units of the nervous system. They are specialized to respond to physical and chemical stimuli, to conduct electrochemical signals, and to release chemicals that regulate various body processes.

The activity of neurons is supported by another type of cells called glial cells. (The word glial comes from a Greek word that means “glue.”) **Glial cells** outnumber neurons by about 10 to 1, and they account for about half of the volume of the nervous system. Collectively, glial cells nourish the neurons, remove their wastes, and defend against infection. Glial cells also provide a supporting framework for all the nervous-system tissue. Figure 11.4 shows a small sample of this tissue.

Individual neurons are organized into tissues called **nerves**. Figure 11.5 shows how hundreds of individual neurons are grouped into nerve bundles and surrounded by protective connective tissue. Like optical fibre cables, which carry many individual wires from one part of a network to another, the nerves extend the neurons throughout the peripheral nervous system. Some nerves consist of neurons that carry information from sensory receptors. Other nerves consist of neurons that carry information to the muscles or glands.

Figure 11.5 Neurons are bundled together into nerves in the peripheral nervous system and into tracts in the central nervous system. Nerves are macroscopic structures. Neurons, however, are microscopic structures. Even a neuron that is over 1 m long cannot be seen with the unaided eye.

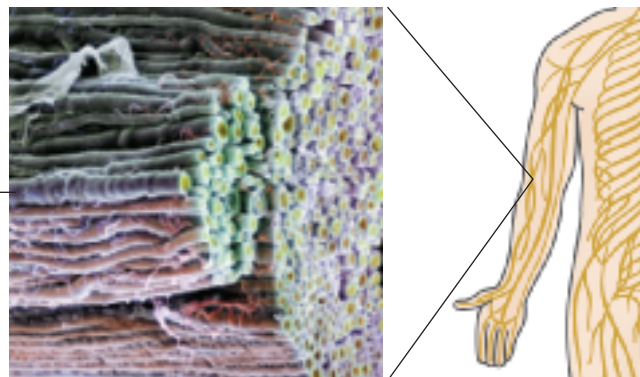


Magnification: 295 x

Figure 11.4 This electron micrograph shows interconnecting neurons with their cell bodies (yellow) and a supporting glial cell (purple).

Neurons vary considerably in size and appearance, depending on their position and function in the body. Three main types of neurons, however, form the basic impulse-transmission pathway of the entire nervous system (Figure 11.6). This pathway, summarized in Table 11.1, depends on three overlapping functions: sensory input, integration, and motor output.

- 1. Sensory input:** Sensory neurons gather information from the sensory receptors (senses) and transmit these impulses to the central nervous system (brain and spinal cord).
- 2. Integration:** Interneurons are found entirely within the central nervous system. They act as a link between the sensory and motor neurons. They process and integrate incoming sensory information, and relay outgoing motor information.



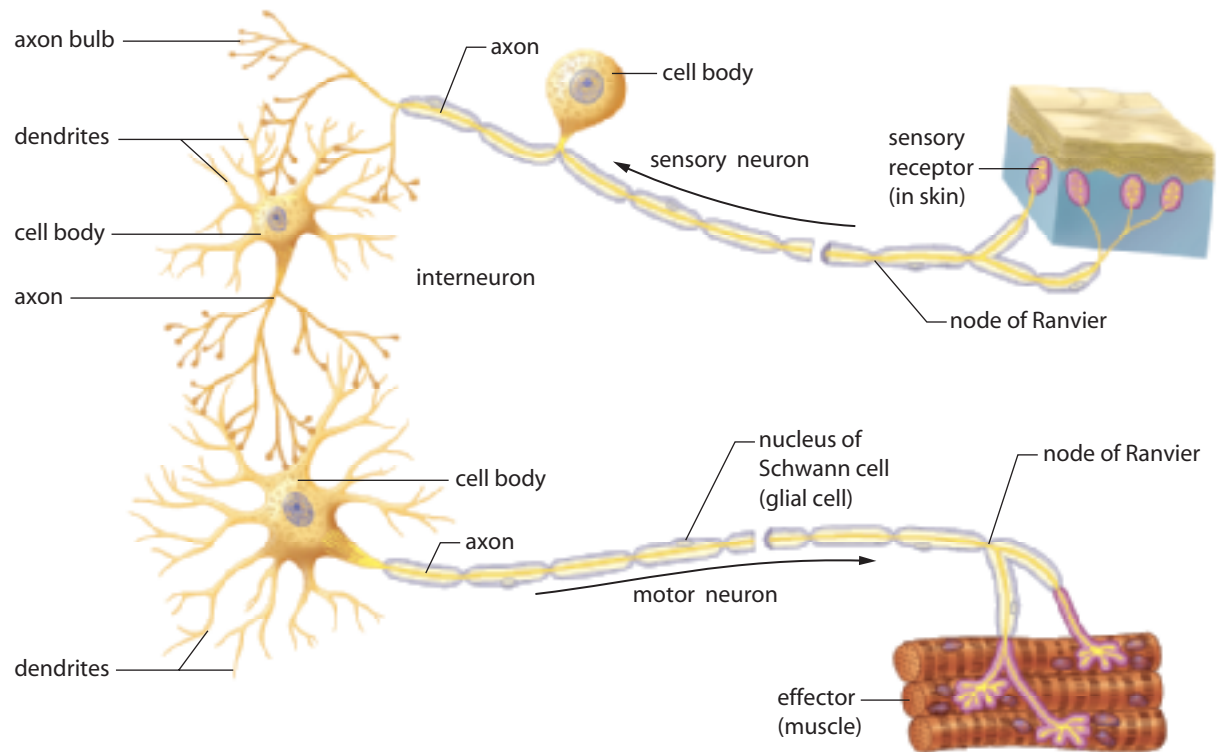


Figure 11.6 This diagram shows how a sensory neuron, an interneuron, and a motor neuron are arranged in the nervous system. (The breaks indicate that the axons are longer than shown).

Table 11.1 Structures of the General Neural Impulse Transmission Pathway

Structure	Function
sensory receptors	receive stimuli and form a nerve impulse
sensory neurons	transmit impulses from the sensory receptors to the interneurons
interneurons	are found in the brain and spinal cord; act as an integration centre
motor neurons	conduct impulses from the interneurons to the effectors
effectors	muscles, glands, and other organs that respond to impulses from the motor neurons

3. Motor output: Motor neurons transmit information from the central nervous system to the muscles, glands, and other organs (effectors).

Figure 11.7 on page 370 provides an example of a basic neural transmission pathway. Suppose that you are driving a car, and a cat darts onto the road in front of you. Your eyes collect sensory information (the sight of the cat on the road), and sensory neurons transmit

this information by conducting electrochemical signals to the brain and spinal cord. Here the information is integrated by interneurons. Motor neurons then carry motor output signals to the muscles (effectors), causing you to extend your foot and press the brake.

The Reflex Arc

Some neurons are organized to enable your body to react rapidly in times of danger, even before you are consciously aware of the threat. These sudden, unlearned, involuntary responses to certain stimuli are called reflexes. Examples of reflexes are jerking your hand away from a hot or sharp object, blinking when an object moves toward your eye, or vomiting in response to food that irritates your stomach. **Reflex arcs** are simple connections of neurons that explain reflexive behaviours. They can be used to model the basic organization of the nervous system.

Reflex arcs use very few neurons to transmit messages. As a result, reflexes can be very rapid, occurring in about 50 ms (milliseconds). Withdrawal reflexes, for example, depend on only

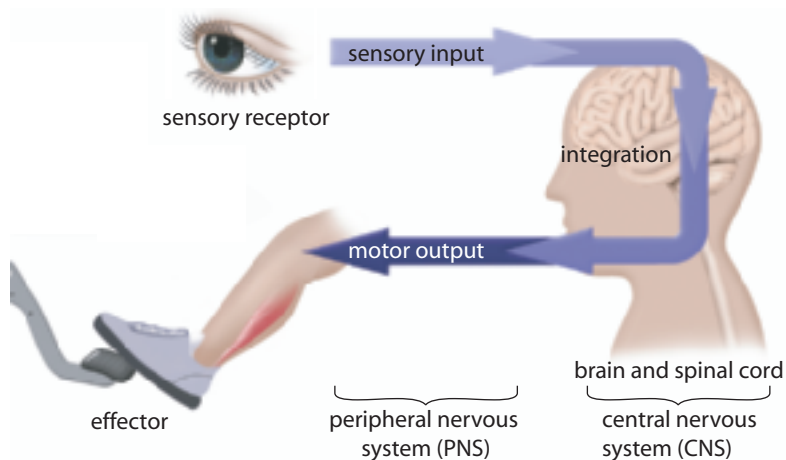


Figure 11.7 An overview of the neurons in a nerve impulse pathway

three neurons. Figure 11.8 illustrates a typical neural circuit, as well as a withdrawal reflex from a potentially dangerous situation. Receptors in the skin sense the pressure of the cactus needle and initiate an impulse in a sensory neuron. The impulse carried by the sensory neuron then activates the interneuron in the spinal cord. The interneuron signals the motor neuron to instruct the muscle to contract and withdraw the hand.

A reflex arc moves directly to and from the brain or spinal cord, before the brain centres involved with voluntary control have time to process the sensory

information. This is why, after stepping on a stone, you would not feel pain or cry out until *after* your foot was withdrawn, once the brain has had time to process the information. You can test your reflexes in Investigation 11.A.

- 4 Compare the basic function of neurons and glial cells.
- 5 List the three types of neurons, and identify their primary functions.
- 6 Identify the basic neural pathway that is involved as you dodge a wayward tennis ball. Compare this pathway with a withdrawal reflex.

The Structure of a Neuron

Neurons have many of the same features as other body cells. In addition, neurons have specialized cell structures that enable them to transmit nerve impulses. Different types of neurons are different shapes and sizes. In general, however, they share four common features: dendrites, a cell body (soma), an axon, and branching ends (Figure 11.9).

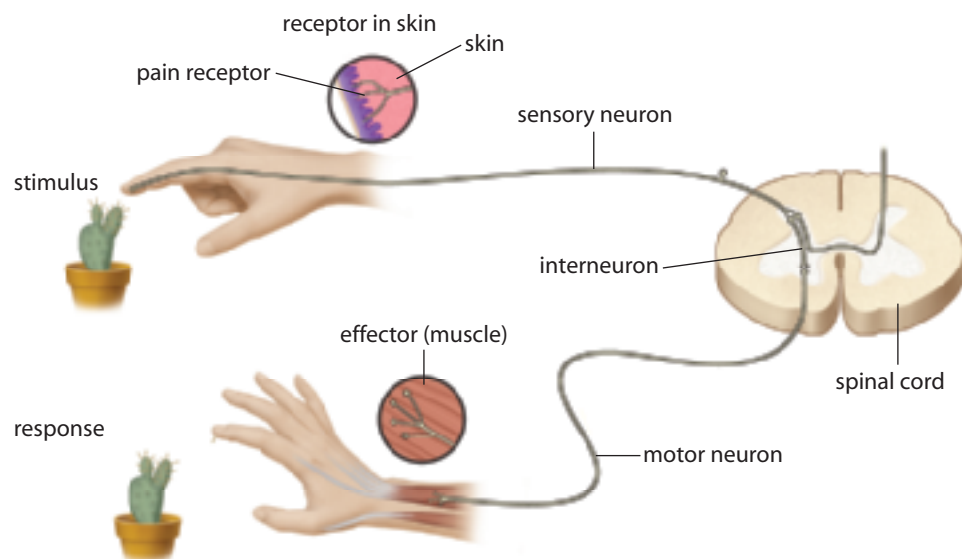


Figure 11.8 A withdrawal reflex. Receptors in the skin perceive the stimulus. Sensory information is conducted from the senses into the spinal cord. Motor information is then conducted away from the spinal cord to the muscles and glands.

Performing an experiment to investigate the physiology of reflex arcs

Identifying a reflex arc not discussed in the textbook, and designing an experiment to investigate the physiology of this reflex arc

Move Fast! Reflex Responses

Reflexes are rapid, involuntary neural pathways that help to protect the body. The knee-jerk (patellar) reflex is an example of the numerous stretch adjustments your body makes every second to unconsciously maintain your balance and coordination. (Your teacher may add others.) Try to initiate the following three common reflexes. Then design an experiment to test a fourth reflex.

Question

How does a reflex arc help to protect the body from harm?

Safety Precautions

Do *not* use excessive force when testing the knee-jerk reflex.

Materials

- cotton balls
- room light
- 20 cm by 20 cm clear plastic sheet
- chair

Procedure

Part 1: Pupillary Reflex

1. Work with a partner. Dim the lights in the room for a few minutes. Look at the pupils in your partner's eyes.
2. Turn on the lights. Check the size of the pupils.

Part 2: Blink Reflex

1. Have your partner hold a piece of clear plastic in front of the face.
2. Without warning, quickly throw a cotton ball at your partner's eyes. Your partner should blink, demonstrating the blink reflex.

Part 3: Knee Jerk (Patellar) Reflex

1. Have your partner sit in a chair with legs crossed, so the top leg can swing freely.
2. Hit the top leg softly, just below the knee, with the side of your hand. The leg should kick out immediately, demonstrating the patellar reflex.

Analysis

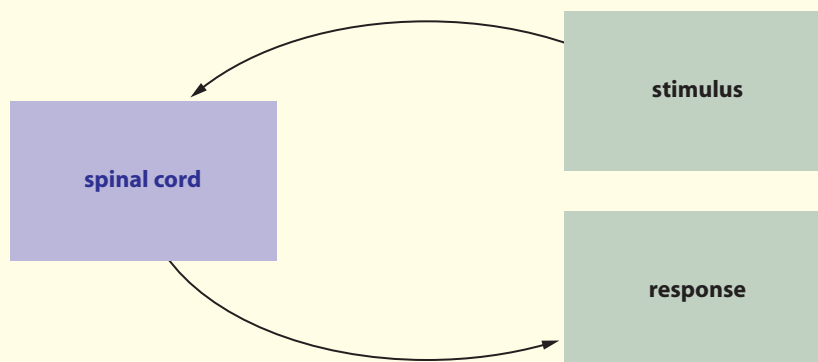
1. Copy the diagram below into your notebook. Use it as a model to represent and summarize each of the reflex responses that you tested and observed in this investigation.
2. Explain why reflexes are important for the body.

Conclusion

3. Describe how the three reflexes tested in this investigation might protect the body.

Applications

4. A student puts her hand on hot glassware, but withdraws her hand *before* she feels the pain. Explain how and why her awareness of the pain is delayed.
5. Design a procedure to test one of the other reflexes of the body. Hypothesize what the reflex arc might look like for the reflex you are testing, and suggest a way in which this reflex would help to protect the body. Obtain your teacher's approval before carrying out your procedure.



BiologyFile

FYI

Researchers at the Max Planck institute have interfaced actual living neurons with microchip transistors, as shown in the photo. They succeeded in sending a signal from a microchip, through two neurons, to another microchip that turned on a silicon switch. They plan to expand their work to understand more about the elaborate communication systems involved in the nerve pathways of the brain.



Dendrites are short, branching terminals that receive nerve impulses from other neurons or sensory receptors, and relay the impulse to the cell body. The dendrites are numerous and highly branched, which increases the surface area available to receive information. The **cell body** contains the nucleus and is the site of the cell's metabolic reactions. The cell body also processes input from the dendrites. If the input received is large enough, the cell body relays it to the axon, where an impulse is initiated.

A neuron typically has one **axon**, which conducts impulses away from the cell body. Axons range in length from 1 mm to 1 m, depending on the neuron's location in the body. For example, the sciatic nerve in the leg contains neuronal axons that extend from the spinal cord all the way to the muscles in the foot, a distance of over 1 m. The terminal end of an axon branches into many fibres. To communicate with adjacent neurons,

glands, or muscles, the axon terminal releases chemical signals into the space between it and the receptors or dendrites of neighbouring cells.

The axons of some neurons are enclosed in a fatty, insulating layer called the **myelin sheath**, which gives the axons a glistening white appearance. These axons are said to be myelinated. Axons without a myelin sheath are said to be unmyelinated (Figure 11.9).

The myelin sheath protects myelinated neurons and speeds the rate of nerve impulse transmission. **Schwann cells**, a type of glial cell, form myelin by wrapping themselves around the axon. In the central nervous system, myelinated neurons form what is known as *white matter*, and unmyelinated neurons form the *grey matter*. Most neurons in the peripheral nervous system are myelinated.

- 7 Draw a neuron, label its basic structures, and identify their functions.
- 8 Describe the structure of a myelinated neuron.

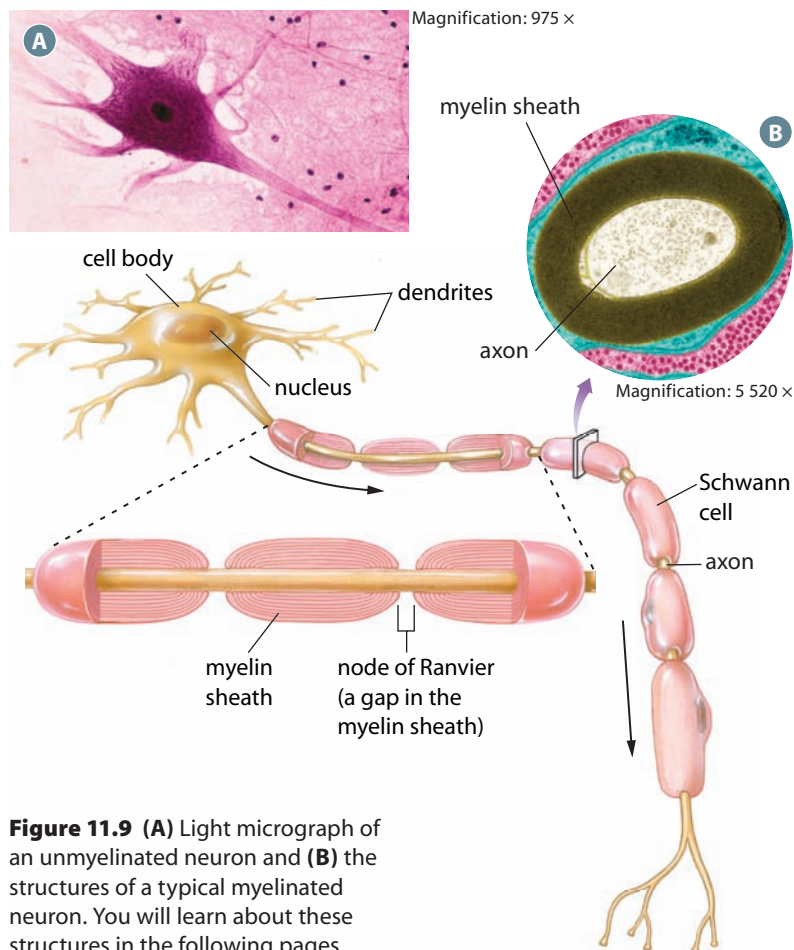


Figure 11.9 (A) Light micrograph of an unmyelinated neuron and (B) the structures of a typical myelinated neuron. You will learn about these structures in the following pages.

Unmyelinated Nerve Impulse

A taser is a non-lethal weapon that some police officers use to immobilize threatening people. A taser sends 25 000 to 150 000 V (volts) of electricity through the human nervous system. This overloads the motor neurons of the somatic system, momentarily incapacitating the body. The electrical jolt interferes with the nervous system because neuron function depends on the cells' ability to conduct electrochemical impulses.

Neurons are able to establish a voltage difference between the inside and outside of the cell membrane. They use this voltage difference to generate a neural impulse. In the remainder of this section, you will focus on how these impulses are generated in neurons and how these electrical signals are transmitted along the length of a neuron.